HARAMAYA UNIVERSITY

COLLEGE OF BUSINESS AND ECONOMICS

DEPARTMENT OF ECONOMICS

Assignment for economics

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Instructions

- Show all the necessary steps.
- Don't copy from each other, if you do that your result will be zero.
- Use your handwriting, computer printed will not allowed
- Submit it within two weeks, (before 17/5/2025)

Questions

1) Find the equilibrium price and quantity of each commodity, when the demand functions of the 3 commodities: A, B and C are given as:

   $DA = 20 - 3PA + PB + PC$

   $DB = 30 + PA - 5PB$

   $DC = 15 + PA - 3PC$ and the corresponding supply functions are

   $SA = 9PA$

   $SB = 30PB - 3$

   $SC = 3PC - 1$

2) Given demand function as $Q = (10 - P^2)^{0.5}$ then;

   A) Derive the function of price elasticity of demand

   B) If p=1birr, calculate Q and price elasticity of demand?

3) Suppose that the firm's production function; TPL= $40L^2 - L^3$
   A) Show how the APL raises when MPL > APL
   B) Show as the APL falls when MPL< APL and
   C) Show as the APL is horizontal when MPL=APL
4) A firm faces the production function $Q=12K^{0.4}L^{0.4}$ and assume it can purchase K & L at prices per unit of 40 birr and 5 birr respectively and it has a budget of 800 birr. Determine the amount of K & L which maximizes output and Compute maximum output produced by firms
5) If $Q=\sqrt{750-10P}$,
   i. What is output maximizing total revenue (TR)
   ii. Calculate market price and total revenue
   iii. If MC=2Q, Calculate maximum profit/minimum loss
6) Consider in perfect competitive market, and then Show that: -
   a. $\text{MR} = P(1-\frac{1}{ed})$
   b. $\text{MR} = AR(1-\frac{1}{ed})$ or $\text{AR} = MR(\frac{ed}{ed-1})$
7) Assume that the function U (x, y) = $X^{0.3}y^{0.5}$ *is* the utility function of a person who consumes two goods in quantities Qx and Qy, respectively. The price of x is Px= 5 and the price of y is Py =8. This person´s income is M = 160.based on this given:
   A. Find the optimal consumption choice of this person
   B. Verify that at the optimum that you found the MRS x y equals the price ratio.
   C. Explain your result in terms of economic theory
   D. Assume that the price of Qx falls to Px= 4. Draw the old and the new budget constraints in a diagram. (Indicate at what values they intersect the axes)
8) Consider a perfectly competitive market in the short run. Assume that market demand is P = 100 - 4 Qd and market supply is P = Qs. And TFC=50, MC=8q+12q²
   A. What is the market equilibrium price and quantity?
   B. Do firms make a profit or loss in the short run, and how much are these profits/losses?
   C. What is the minimum value of MC and AVC?

D. How much should be the minimum price of product firm needed to stay in market?

9) Derive the total cost (TC) function using the production function: $\mathbf{Q} = \mathbf{L}^{\frac{2}{3}}\ \mathbf{K}^{\frac{1}{3}}$ and price of labor (w) =2 and price of capital (r) =4.

10) Suppose a certain contractor wants to maximize profit ($\prod$) from building one bridge. The contractor uses efficient combination of both labor and capital that is sufficient to make a bridge indicated by the function $\mathbf{0.25\ L^{0.5}\ K^{0.5}}$. If the prices of labor (w) and capital (r) are Birr 5 and Birr 10 respectively. Calculate the efficient combinations of labor(L) capital(K) and minimum cost (C) of the production.

11) Mr. Haptish`s preferences for tea and coffee is given by **U (X, Y) = 2x + y,** where X denotes the units of tea and Y is units of coffee. His income is Birr 500, and prices for Px = Birr 36 and Py = Birr 22, given this all, what is his utility-maximizing pair of tea and coffee? Calculate his maximum utility!

12) Suppose a firm has a TFC of Birr 2,000, a TVC of Birr 5,000 and a TR of Birr 6,000 at equilibrium. Should the firm stop its operation? Why?

13) Suppose you are given the following information about a particular industry: Market demand **Qd = 6500 - 100P** and market supply **QS = 1200P**, Firm total cost function $\mathbf{C(q) = 722 + \frac{q^2}{200}}$ Assume that all firms are identical and that the market is characterized by perfect competition and then based on the given data: -

- **A.** Find the equilibrium price and the equilibrium quantity
- **B.** Calculate the output supplied by the firm, and the profit of each firm
- **C.** Find the lowest price at which each firm would sell its output in the long run?
- **D.** Calculate and explain whether the profit is positive, negative, or zero at this price (price you get on C)
- **E.** Would you expect to see entry into or exit from the industry in the long run? Explain. What effect will entry or exit have on market equilibrium?